Sri Sivasubramaniya Nadar College of Engineering

(An Autonomous Institution affiliated to Anna University)
Rajiv Gandhi Salai, Kalavakkam-603110

Stimuli for Technological Innovation and Research among Students (STIRS) - 2026

Category

☐ Research Project ☐ Innovative Product Development

Project/Product Title

Closed-Loop Control and Dynamic Phase Shedding for a Two-Stage Multiphase DC-DC Converter in EV Auxiliary Power Applications

NAME	YEAR	DEPARTMENT
Boobesh IR	3rd Year	EEE
Deepalakshmi M	3rd year	EEE

Guided by
Dr. Seyezhai R
Professor, Department of electrical and Electronics Engineering

Budget: Rs.15,000/-

Project Duration :12 months

Signature of the Project Students Signature of the Project Guide(s)

Signature of the HOD

Annexure A
(Applicable for Research Project only)

1. Project Title:

Closed-Loop Control and Dynamic Phase Shedding for a Two-Stage Multiphase DC-DC Converter in EV Auxiliary Power Applications

2. Broad Subject:

Power Electronics / Electric Vehicle Auxiliary DC-DC Converters / Digital Control of Multiphase Converters

3. Project Duration *(in months)*:

12 Months

4. Budget: Rs.15,000/-

5. Project Summary:

Electric vehicle (EV) low-voltage auxiliary rails (12 V / 24 V / 48 V) demand efficient DC-DC conversion under widely varying load. Multiphase interleaved converters reduce ripple and device stress, while phase shedding improves light-load efficiency by deactivating unused phases. Two-stage non-inverting architectures (for example, interleaved buck with a shared boost) support dual-mode operation relevant to EV auxiliary and intermediate-rail applications; however, published work largely separates these themes: shared-stage interleaved topologies are often demonstrated without adaptive phase shedding, and phase-shedding control with current sharing is typically validated only on single-stage multiphase bucks. Building on an existing open-loop simulation proof-of-concept of a two-stage parallel interleaved converter (two-phase interleaved buck + shared boost) with Monte Carlo analysis, this project implements closed-loop voltage regulation and inter-phase current sharing integrated with dynamic phase shedding, and validates a low-power hardware prototype using an STM32 digital controller. Expected outcomes include measured efficiency versus load (with and without shedding), phase-current balance, load-step and phase add/drop transients, and a documented comparison of open-loop versus closed-loop shedding performance suitable for publication and further proposals.

6. Keywords:

Multiphase interleaved DC-DC; phase shedding; EV auxiliary loads; closed-loop control; current sharing; two-stage converter; STM32; light-load efficiency

7. Objectives:

1. Design and simulate a two-stage multiphase DC-DC converter for EV-auxiliary voltage class (primary demo: 48 V to 12/24 V buck path; dual-mode analysis including 24 V to 48 V boost) with outer voltage regulation and inter-phase current sharing.

2. Integrate load-adaptive dynamic phase shedding (add/drop thresholds with hysteresis and dwell) with the closed-loop controller; quantify regulation, current-share error, and efficiency versus an open-loop shedding baseline on the same plant.

3. Implement the control algorithm on an STM32 platform (multiphase PWM generation, ADC sensing, real-time shed/add state machine) and verify in firmware tests and power-stage bench experiments.

4. Fabricate and test a low-power prototype (approximately 200–300 W class), reporting measured efficiency curves, phase current balance, thermal/current stress under shedding, and load-step / phase-transition waveforms.

5. (Stretch) Analyse multiphase boost-side extension (for example, 4-phase buck + 2-phase boost) to restore phase-shedding authority in boost mode; simulate and document trade-offs versus the baseline shared-boost topology.

8. Introduction:

The growth of battery electric vehicles has increased both high-voltage traction demand and low-voltage auxiliary demand (lighting, ECUs, sensors, actuators, and 48 V loads such as compressors and power steering). Industry surveys of auxiliary power modules (APMs) emphasise efficient conversion among 12 V, 24 V and 48 V networks, higher power density, and in some cases reverse power capability for start/backup functions. Within this landscape, multiphase interleaved DC-DC converters are attractive because interleaving cancels ripple, distributes thermal stress, and supports modular current rating. At partial load, common for auxiliaries, keeping all phases active increases switching and gate-drive losses; phase shedding (phase dropping) is a standard remedy on multiphase bucks. Two-stage non-inverting topologies enable buck and boost capability without inverted polarity and can reduce device voltage stress relative to single-switch buck-boost. Prior interleaved converters and composite automotive architectures motivate multi-stage efficiency, while recent digital multiphase work shows that closed-loop current control can be combined with rotating phase shedding for thermal balance. This project focuses on the control and experimental gap that remains when those ideas meet a shared-stage two-stage plant for EV-auxiliary voltage levels: how to regulate voltage and share current while dynamically changing the active phase set, and how to validate the resulting efficiency and transient behaviour in hardware.

9. Definition of the Problem:

n a two-stage converter architecture combining a multiphase power stage with a shared, single-leg companion stage, simulation results show that open-loop phase shedding can boost light-load efficiency—but it offers no guarantees around output voltage regulation, current sharing across phases, or smooth transient behavior during phase addition or removal. Because individual phase components inevitably differ in inductance, on-resistance, and current-sense gain, the phases that remain active after shedding tend to absorb a disproportionate share of thermal stress. A further limitation arises in boost mode: since the companion stage is a single leg, it lacks any phase-shedding flexibility of its own, which constrains how much dual-mode (buck/boost) efficiency optimization is achievable.

Existing work covers several related pieces but not their integration. Multiphase non-inverting converter studies tend to implement minimal or no dynamic phase-shedding capability.

Separately, multiphase buck and battery-charger designs demonstrate phase shedding paired with closed-loop current regulation. More recent work on fuel-cell powertrain applications explores phase shedding specifically in boost-mode operation. What's missing is a single study that brings these threads together: closed-loop regulation combined with adaptive shedding, validated on hardware, applied to a two-stage plant sized for EV auxiliary-power duty, and benchmarked against an open-loop baseline. This work targets that specific gap. It does not extend into full high-voltage isolated multiport auxiliary power modules or traction-inverter-class systems.

10. Review of status of Research and Development in the subject

10.1 National Status

In India, multiphase and bidirectional DC-DC work for EV and battery systems appears in national conferences and academic laboratories, including multiphase bidirectional converters with load-based phase shedding for drive/regeneration scenarios, and non-inverting buck-boost design studies for battery charging. Much of this work is simulation-centric, fixed-phase, or single-stage, and does not jointly treat two-stage shared-stage plants with closed-loop current sharing and experimental phase-shedding validation for auxiliary rails. Department-level power electronics laboratories typically provide MATLAB/Simulink, programmable supplies, and oscilloscopes suitable for a low-power multiphase prototype, aligned with this proposal's execution plan.

Effective phase utilization and efficiency improvement of high power interleaved DC–DC converter using modified rotating phase shedding control by **Guttula Yedukondalu and Susovon Samanta** (2023, Computers & Electrical Engineering, ScienceDirect — NIT Rourkela, India) proposes a modified rotating phase-shedding algorithm combined with adaptive current-limit control that automatically selects the active phase count based on efficiency estimation, distributing switching stress across all phases to counter the aging and thermal-instability problems of conventional phase-shedding.

Multiphase Bidirectional DC-DC Converter with Phase Shedding for BLDC Driven Electric Vehicle by **S. Ramya, S. Divya, and C. Das** (2021, IEEE IPRECON, Kollam, India) adapts the number of active phases in a bidirectional multiphase converter to the instantaneous load current in a battery-powered EV, analyzing an interleaved soft-switched boost stage to cut switching losses while supplying a BLDC motor load and supporting regenerative-braking charge-back.

Modeling and Control Implementation of Interleaved Coupled and Uncoupled Boost Converter for EV Drive Applications by **Guttula Yedukondalu and Susovon Samanta** (2021, IEEE INDICON, Guwahati, India) compares closed-loop control implementations for coupled- and uncoupled-inductor interleaved boost converters intended for EV drivetrain power conversion, evaluating their dynamic response and current-sharing behavior under load transients.

Energy-efficient and reliable dual closed-loop DC control system for intelligent electric vehicle charging infrastructure by **Jun Li, Wan Chen, Xiaoqiong Zhu, Baoguo Zang, Cong Zhang, Hengxiao Hu, Ming Zhang, and Wenbao Lei** (2024, PLOS ONE — co-authored with SRM Institute of Science and Technology, India) develops a dual closed-loop control scheme (inner current loop, outer voltage loop) for a multilevel PWM rectifier front-end feeding an EV charging system, demonstrating high power factor, low THD, and stable phase-coupled power sharing.

10.2 International Status

Internationally, cascaded multiphase non-inverting interleaved converters have been analysed and experimentally demonstrated (for example, shared buck with n-phase boosts), primarily without adaptive phase shedding and often in charger or discontinuous-mode contexts. Cascaded interleaved bidirectional converters for charging stations employ dual-stage PI-type control with fixed phase count. Multiphase phase shedding for light-load efficiency and ripple management is mature for interleaved bucks and EV fast-charger modules; digital average-current predictive control with rotating phase shedding has been hardware-validated for multiphase bucks. Stacked interleaved boost phase shedding with ripple-free modulation has been reported for fuel-cell powertrains (2025). EV auxiliary converter reviews (APM surveys) stress 12/24/48 V rails, density, and control, but leave multiphase dynamic shedding on two-stage aux-class prototypes under-explored. This proposal positions against that international backdrop.

A Multiphase Buck Converter With a Rotating Phase-Shedding Scheme for Efficient Light-Load Control by Y. Ahn, I. Jeon, and J. Roh (2014, IEEE Journal of Solid-State Circuits) presents a four-phase buck converter that rotates which phases are shed rather than always dropping the same ones, spreading switching-related aging and thermal stress evenly across all phases instead of concentrating it on a single leg.

A Novel Phase-Shedding Control Scheme for Improved Light Load Efficiency of Multiphase Interleaved DC–DC Converters by **F. Su and W.-S. Liu** (2013, IEEE Transactions on Power Electronics) proposes a lookup-table-based efficiency estimation method that automatically selects the optimal number of active phases as load current changes, validated experimentally on a three-phase synchronous buck converter with closed-loop current control.

Current Ripple Minimisation Based on Phase-Shedding of DC-DC Interleaved Converters for EV Charging System by **M. Alharbi, M. Dahidah, S. Ali, S. Ethni, and V. Pickert** (2019, IEEE IECON) introduces a phase-shedding control scheme for a multiphase interleaved buck converter used in EV fast-charging applications, dynamically varying the active phase count across the full output-voltage range to jointly minimize output current ripple and maximize efficiency.

Phase Shedding of a Ripple-Free Stacked Interleaved Boost Converter for Fuel Cell-Based Powertrains (2025, IEEE Xplore) examines a stacked interleaved boost topology for fuel-cell electric vehicles, showing that adding a dedicated fractional leg to a conventional interleaved

boost preserves ripple-free input current under phase shedding while extending efficiency gains into partial-load boost-mode operation.

11. Novelty / Importance of the proposed project in the context of current status

1) Integrated dual-loop control (output voltage regulation + inter-phase current sharing) co-designed with load-adaptive phase shedding on a two-stage EV-auxiliary multiphase converter, rather than open-loop duty maps alone.

2) Explicit open-loop versus closed-loop phase-shedding performance comparison (regulation error, share error, load-step recovery, efficiency) on the same two-stage plant—motivated by observed light-load open-loop voltage deviation in the existing simulation baseline.
3) Hardware validation of efficiency, current sharing, and phase add/drop behaviour on a low-power STM32-controlled prototype, including attention to thermal/current stress on remaining active phases and the shared-stage hotspot.

4) (Stretch) Analysis of dual multiphase extension (multiphase buck + multiphase boost) to restore shedding authority in boost mode, addressing a structural limit of shared-boost two-stage topologies.

Cost-effectiveness: a student-scale silicon MOSFET prototype and open MCU control (STM32) keep hardware costs low while producing transferable control algorithms. Scalability: control and shedding logic scale to higher phase counts; the power stage can be re-rated later. Impact: supports more efficient EV auxiliary conversion and trains manpower in digital power control for automotive power electronics.

12. Patent details *(domestic and international),* if applicable -Nil

13. Work plan and Methodology

Month 1–2: Simulation

- Design and model the proposed two-stage multiphase DC-DC converter in MATLAB/Simulink.
- Verify basic converter operation under different load conditions.

Month 3–4: Closed-Loop Control Development

- Develop the closed-loop voltage controller and current-sharing algorithm.
- Integrate dynamic phase shedding and evaluate converter performance through simulation.

Month 5–6: STM32 Implementation

- Implement the control algorithm on an STM32 microcontroller.
- Configure PWM generation, ADC sensing, and phase-shedding logic.

Month 7–8: Hardware Development

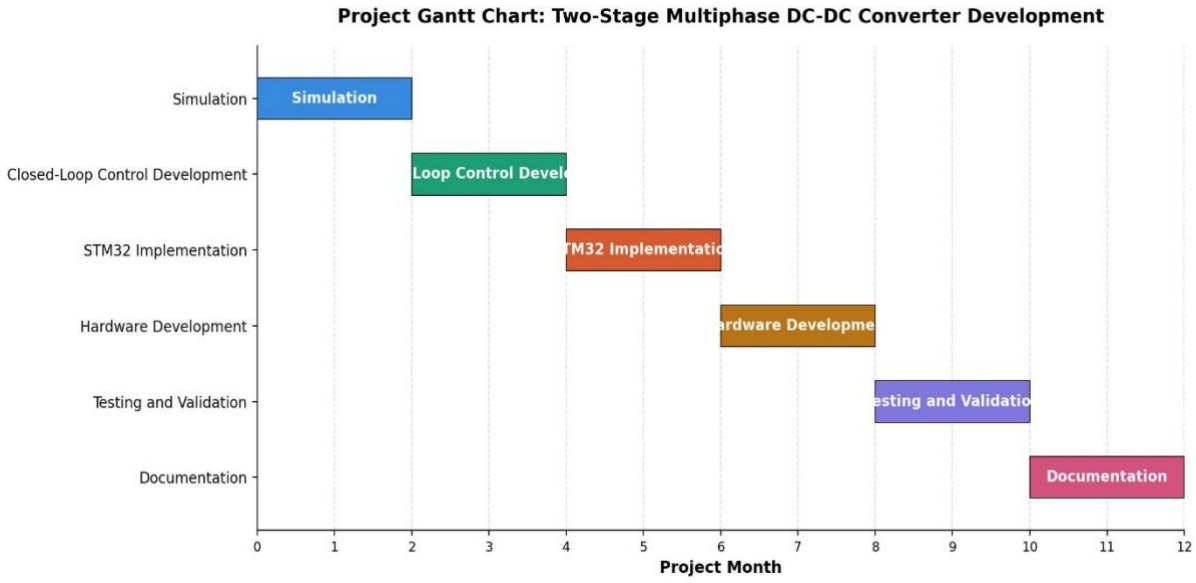
- Fabricate the converter prototype.
- Integrate the STM32 controller, power stage, sensing circuits, and gate drivers.

Month 9–10: Testing and Validation

- Test the hardware under different operating conditions.
- Compare the hardware results with the MATLAB/Simulink results and evaluate efficiency, ripple, voltage regulation, and current sharing.

Month 11–12: Documentation

- Analyze the experimental results.
- Prepare the final project report and research paper for publication.



14. Testing and Validation Strategy

- *Simulation validation:* Compare closed-loop + shedding simulation results against the existing open-loop Monte Carlo baseline for regulation accuracy and efficiency.
- *Bench testing:* Measure efficiency-vs-load curves (with and without shedding), inter-phase current balance, and thermal/current stress on active phases using a programmable DC source, electronic load, and DSO.
- *Transient validation:* Capture load-step and phase add/drop transient waveforms to verify controller stability and recovery time.

- *Baseline comparison:* Benchmark hardware results against the open-loop shedding baseline and against published multiphase phase-shedding efficiency figures to contextualize improvement.

15. Time schedule

Activity	1–2	3–4	5–6	7–8	9–10	11–12
MATLAB/Simulink Modelling	✓					
Closed-Loop Controller Design		✓				
STM32 Implementation			✓			
Hardware Development				✓		
Testing & Validation					✓	
Documentation						✓

16. Deliverables

- Improved converter efficiency, particularly under light-load conditions through dynamic phase shedding.
- Reduced output voltage and current ripple compared to the baseline converter.
- Reduced switching and conduction losses, resulting in improved overall converter performance.
- Stable closed-loop output voltage regulation under varying load conditions.
- Performance comparison between open-loop and closed-loop converter operation.
- MATLAB/Simulink model and STM32-based hardware prototype.
- Experimental waveforms and efficiency plots validating the proposed system.

17. Target beneficiaries of the proposed work

- Electric Vehicle manufacturers
- Automotive power electronics researchers
- Power converter designers
- Industries developing high-efficiency auxiliary power converters

18. Plan for Research Outcome

The outcomes of this project are expected to contribute to both academic research and practical applications.

- Publication of research findings in a reputed IEEE journal or conference.
- Presentation of the work at national or international conferences.
- Evaluation of patentability after completion of hardware validation.
- Further development into a higher-power EV auxiliary converter platform.

19. References:

1. B. N. Alajmi, M. I. Marei, I. Abdelsalam and N. A. Ahmed, "Multiphase Interleaved Converter Based on Cascaded Non-Inverting Buck-Boost Converter," in *IEEE Access*, vol. 10, pp. 42497-42506, 2022, doi: 10.1109/ACCESS.2022.3168389.
2. B. N. Alajmi, I. Abdelsalam, M. I. Marei and N. A. Ahmed, "Two Stage single-phase EV On-Board Charger Based On Interleaved Cascaded Non-Inverting Buck-Boost Converter," *2023 IEEE Conference on Power Electronics and Renewable Energy (CPERE)*, Luxor, Egypt, 2023, pp. 1-6, doi: 10.1109/CPERE56564.2023.10119584.
3. H. Chen, H. Kim, R. Erickson and D. Maksimović, "Electrified Automotive Powertrain Architecture Using Composite DC–DC Converters," in *IEEE Transactions on Power Electronics*, vol. 32, no. 1, pp. 98-116, Jan. 2017, doi: 10.1109/TPEL.2016.2533347.
4. W. C. Leal, M. O. Godinho, R. F. Bastos, C. R. de Aguiar, G. H. F. Fuzato and R. Q. Machado, "Cascaded Interleaved DC–DC Converter for a Bidirectional Electric Vehicle Charging Station," in *IEEE Transactions on Industrial Electronics*, vol. 71, no. 4, pp. 3708-3717, April 2024, doi: 10.1109/TIE.2023.3273281.
5. G. Mao, G. Zhou, H. Zhao, Y. Gao, P. P. Kubulus and S. Munk-Nielsen, "Digital Average Current Predictive Control for Multiphase Interleaved DC–DC Converters," in *IEEE Transactions on Circuits and Systems II: Express Briefs*, vol. 70, no. 12, pp. 4519-4523, Dec. 2023, doi: 10.1109/TCSII.2023.3294760.
6. M. A. Alharbi, M. Dahidah, S. Ethni, S. Ali, N. B. Kadandani and V. Pickert, "A Phase Shedding Control Technique for Ripple Minimisation for EV Fast DC Charging Applications," *2021 23rd European Conference on Power Electronics and Applications (EPE'21 ECCE Europe)*, Ghent, Belgium, 2021, pp. 1-10, doi: 10.23919/EPE21ECCEEurope50061.2021.9570657.
7. S. Ramya, S. Divya and C. Das, "Multiphase Bidirectional DC-DC Converter with Phase shedding for BLDC Driven Electric Vehicle," *2021 IEEE International Power and Renewable Energy Conference (IPRECON)*, Kollam, India, 2021, pp. 1-6, doi: 10.1109/IPRECON52453.2021.9640997.
8. A. V. Rao and G. K.P., "Analysis, Design and Simulation of Non-Inverting Buck-Boost DC-DC Converter for Battery Charging," *2021 International Conference on Disruptive Technologies for Multi-Disciplinary Research and Applications (CENTCON)*, Bengaluru, India, 2021, pp. 79-84, doi: 10.1109/CENTCON52345.2021.9688165.
9. G. J. Sagar, V. V. Nag, T. Mahto and P. Mishra, "A High-Efficiency EV Charging System with Interleaved Buck-Boost Converter and Adaptive Control," *2025 IEEE North-East India International Energy Conversion Conference and Exhibition (NE-IECCE)*, Silchar, India, 2025, pp. 1-6, doi: 10.1109/NE-IECCE64154.2025.11183500.
10. P. Pilati, R. Mandrioli, V. Cirimele, D. Guilbert and M. Ricco, "Phase Shedding of a Ripple-Free Stacked Interleaved Boost Converter for Fuel Cell-Based Powertrains," in *IEEE Transactions on Transportation Electrification*, vol. 11, no. 6, pp. 13999-14008, Dec. 2025, doi: 10.1109/TTE.2025.3609934.

20. List of facilities and Equipment available with Department for the project

S.No	**Equipment Name / Funding Source**	**For which purpose it would be utilized in current project**
1	Scope Corder, DL850 (MODEL NO: D L850FHE) Make; Yogokowa	To Trace the output waveforms simultaneously with different channels
2	Single-phase PQ Analyzer -Fluke	Used to Measure different parameters such as power quality, THD, Current a nd Voltage ripples
3	Digital Storage Oscilloscope, DSO (Scientific Instruments)	Trace the output Voltage and current waveforms
4	DSPIC Controller	To Generate the Gate Pulse for Power switches
5	LCR Meter, Digital Multi Meter	To measure different inductive and ca pacitive components

21. Budget Estimates

S.No	**Components**	**Rating**	**Unit Price**	**Quantity**	**Total Cost (Rs.)**
1	Power MOSFET (with heatsink tab)	IRFP4568 / 150V, 34A	150	8	1,200
2	Gate driver IC& Protection	IR2110 / IRS2110	215	4	860
3	Buck-stage phase inductors	100 µH / 10A	350	4	1,400
4	Boost-stage phase inductors	220 µH / 6A	300	2	600
5	Fast recovery / Schottky diodes	MBR20100CT / MUR460	50	6	300
6	Output filter capacitors	100 µF / 100V	50	6	300
7	Input filter capacitors	470 µF / 63V	60	4	240
8	Current sense modules (per-phase, for current sharing)	ACS712 / shunt + amp	150	6	900
9	STM32 Development controller board	STM32 (multiphase PWM + ADC)	3,200	1	3,200
10	Custom power-stage PCB fabrication	2-layer, prototype run	2,500	1	2,500
11	Heat sinks, connecting wires, snubbers, patch chords, banana connectors (M&F)	—	100	20 nos each	2,000

12	Miscellaneous (fuses, fasteners, connectors)				500
13	Fibre enclosure sheet, spacers & standoffs	A4 Size	500	2	1000
				Total	**15,000/-**

22. Budget Justification

Components	**Justification**
MOSFET device module fast recovery diodes, inductors & capacitors	To construct the power circuit of the proposed converter
Digital Processor& gate drive circuits	For pulse generation & interface
Connecting wires, Heat sinks, Protection circuits, patch chords, banana connectors (male & female)	For building the power circuit
Multimeter	To measure the voltage and current values.
connecting wires, patch cords & spacers	For interconnections